

Mixture of Concept Bottleneck Experts

Francesco De Santis, Gabriele Ciravegna, Giovanni De Felice, Arianna Casanova, Francesco Giannini, Michelangelo Diligenti, Johannes Schneider, Danilo Giordano, Mateo Espinosa Zarlenga, Pietro Barbiero
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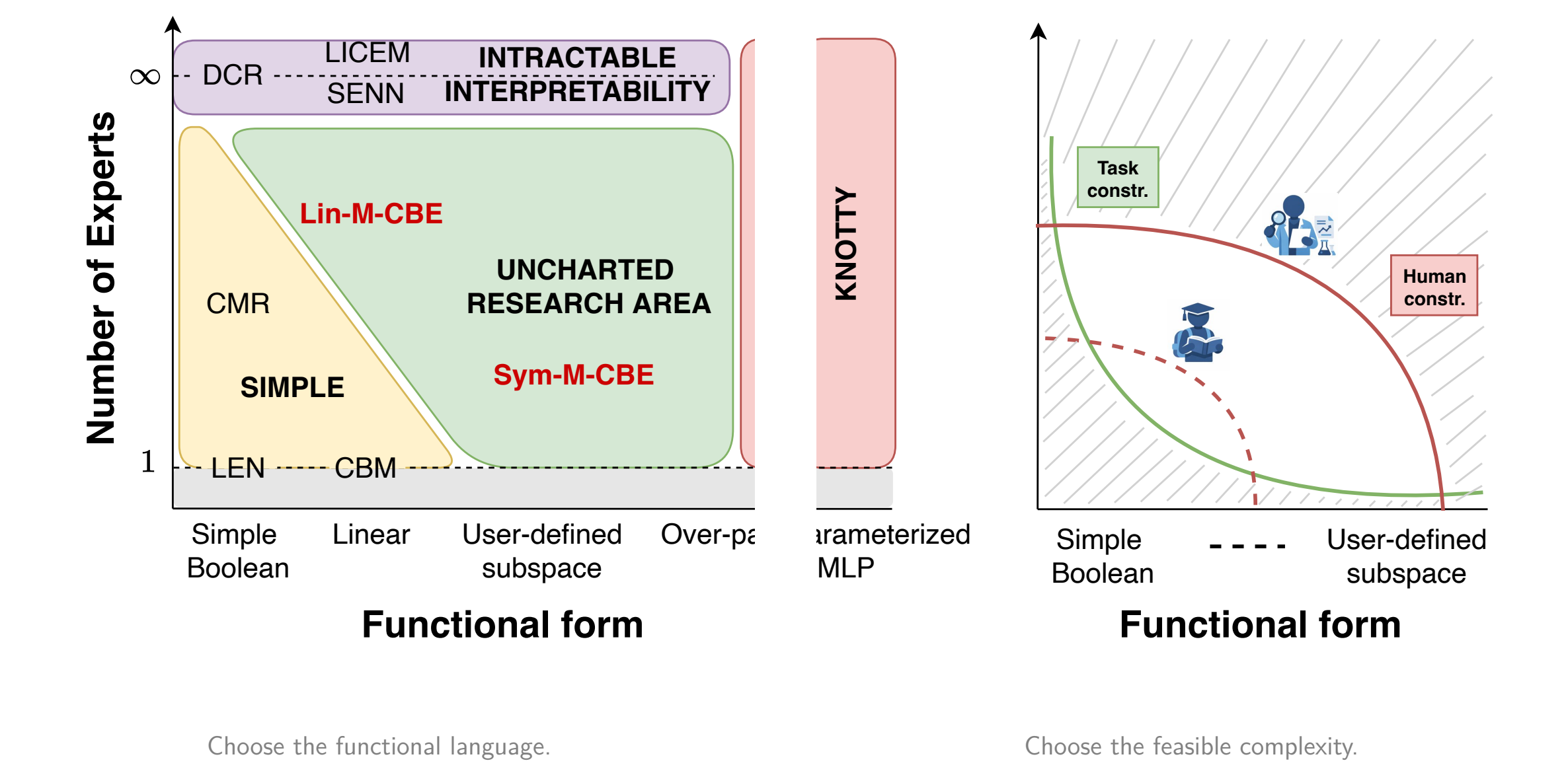
Why

Concept-based models can tell us **which human variables** matter, but often hide or over-fix **the mechanism** that turns those variables into a prediction.

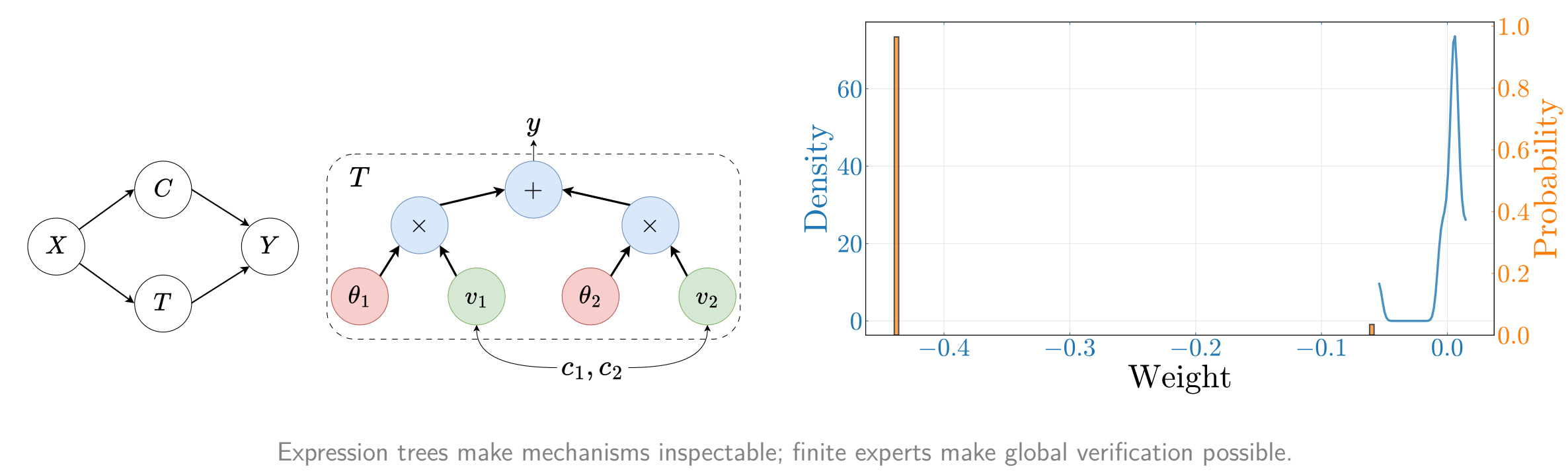
Core Idea

M-CBE learns a **finite menu of readable mechanisms**. Each prediction selects one mechanism, and each mechanism uses only predicted concepts.

Two Axes of Control



Interpretability



Instances

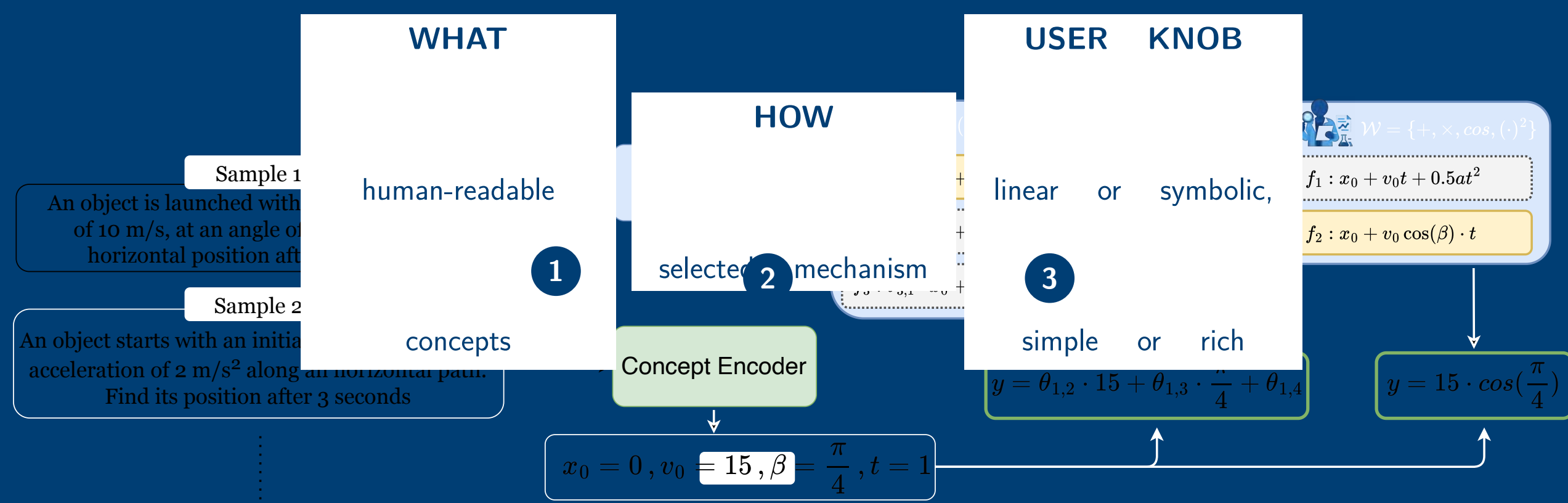
Lin-M-CBE: multiple linear mechanisms.

Sym-M-CBE: symbolic mechanisms under user-chosen operators.

Explain what the model sees and how it reasons.

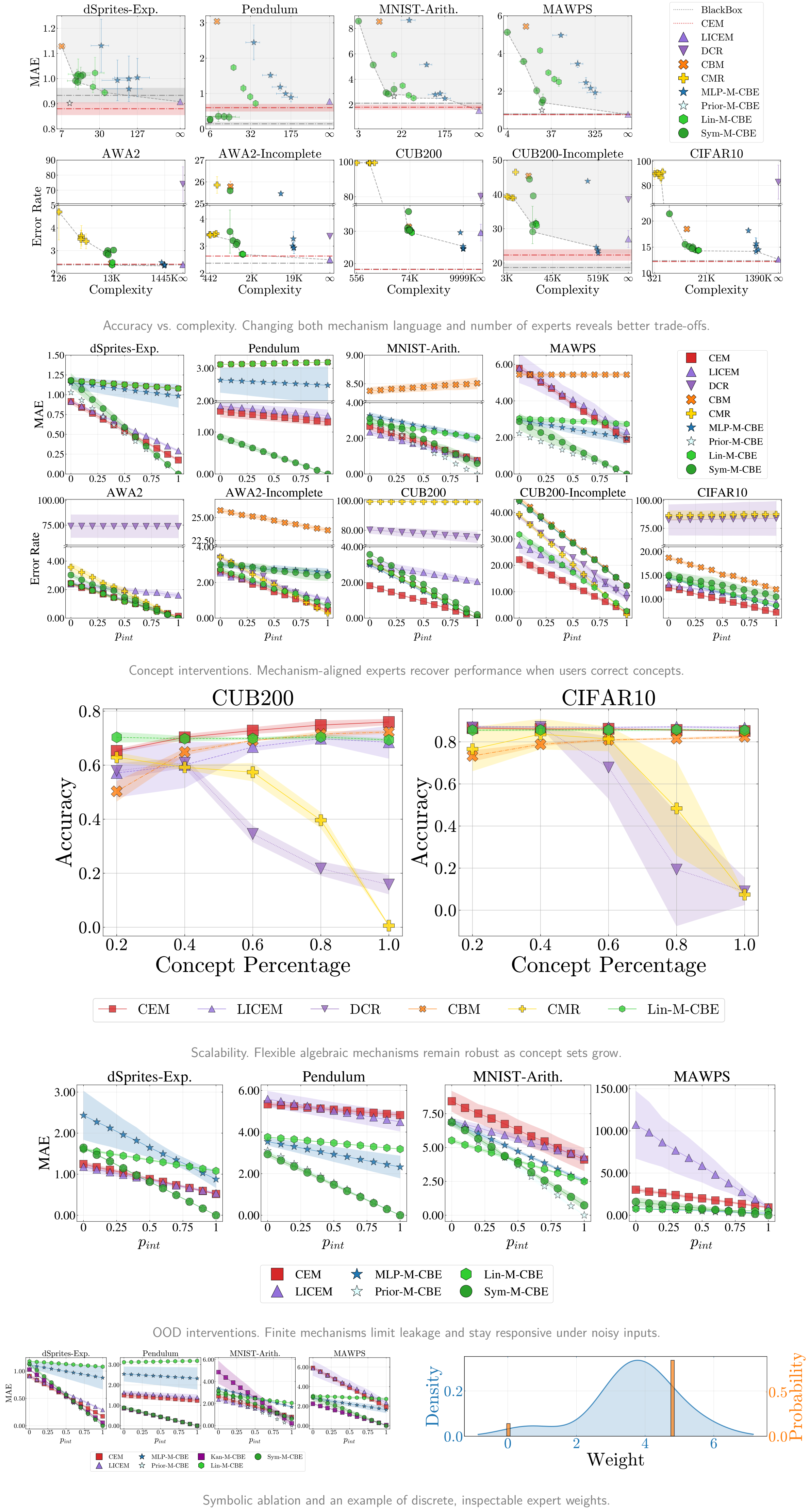
Human concepts + readable mechanisms,

at the complexity users choose.



Experiments

Nine benchmarks: regression and classification, continuous and discrete concepts, complete and incomplete bottlenecks, plus label-free concepts.



Mixture of Concept Bottleneck Experts exposes both the variables and the mechanisms behind a prediction.



Code and experiments